

BIG RAPIDS ROBEN-HOOD, MI, USAWMO: **725416**Lat: **43.717N**Lon: **85.500W**Elev: **283**StdP: **97.97**Time Zone: **-5.00 (NAE)**Period: **99-19**WBAN: **14864****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.9	-17.3	-23.4	0.5	-20.2	-20.2	0.6	-15.7	9.3	-4.3	8.5	-4.6	0.6	270	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	31.6	21.6	29.8	21.0	27.9	20.2	23.7	28.3	22.7	27.3	21.8	26.1	4.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	17.7	26.1	21.3	16.5	25.0	20.5	15.7	24.3	72.1	28.5	68.4	27.3	65.3	26.0	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.7	6.9	-26.0	33.5	4.0	1.8	-28.9	34.8	-31.2	35.8	-33.4	36.9	-36.3	38.2		
(5)			-26.3	25.4	3.9	1.4	-29.1	26.3	-31.4	27.1	-33.6	27.9	-36.5	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.8	-5.7	-5.3	0.0	6.9	13.0	18.3	20.6	19.5	15.6	9.1	3.1	-2.6
	DBStd	10.54	6.10	5.94	6.29	5.04	4.65	3.77	3.41	3.16	4.18	4.73	4.82	5.46
	HDD10.0	2057	488	429	316	120	21	0	0	0	4	77	212	391
	HDD18.3	4081	746	662	569	345	176	47	16	23	103	288	457	649
	CDD10.0	1247	0	0	6	26	114	249	328	296	172	49	6	1
	CDD18.3	228	0	0	1	1	11	46	86	61	20	2	0	0
	CDH23.3	2519	0	0	5	25	153	531	943	612	229	21	0	0
	CDH26.7	714	0	0	1	4	33	162	302	157	53	2	0	0
Wind	WSAvg	2.6	2.9	3.0	2.9	3.2	2.8	2.4	2.1	2.0	2.1	2.5	2.8	2.9
Precipitation	PrecAvg	844	52	38	58	75	80	80	67	96	94	72	76	55
	PrecMax	1170	102	107	142	144	220	244	204	288	484	196	181	134
	PrecMin	518	15	7	8	15	16	20	14	10	1	12	11	12
	PrecStd	153	24	23	31	30	47	44	39	58	74	37	39	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.4	12.9	23.1	26.3	29.8	32.6	33.9	32.6	31.2	26.1	20.0	13.7
		MCWB	8.6	8.7	17.1	15.6	19.7	22.7	22.8	21.7	20.6	19.1	13.7	11.5
	2%	DB	7.2	7.9	16.4	22.8	27.4	30.9	32.1	30.1	28.0	22.8	16.2	8.9
		MCWB	4.8	5.6	11.3	13.5	18.5	21.2	22.0	21.3	20.1	16.5	11.6	7.4
	5%	DB	4.0	5.1	12.8	20.1	24.8	28.4	29.9	28.0	26.2	20.2	13.6	6.7
		MCWB	2.6	2.8	8.9	12.1	16.8	20.3	21.4	20.4	19.1	16.1	10.0	4.9
	10%	DB	2.3	2.6	9.4	17.1	22.4	26.4	27.9	27.0	23.9	17.7	11.3	4.1
		MCWB	1.4	1.0	5.8	10.2	15.5	18.8	20.4	19.7	18.1	14.0	8.5	2.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	10.2	16.9	18.0	21.7	24.1	25.7	24.4	23.3	21.2	15.0	12.2
		MCDB	9.7	11.4	22.4	24.0	26.6	29.7	31.0	28.9	26.8	24.9	17.8	13.4
	2%	WB	4.9	6.1	13.0	15.3	20.0	22.6	24.1	23.1	21.8	18.7	12.6	7.9
		MCDB	6.5	8.3	15.7	21.0	24.7	28.1	28.7	27.5	25.5	20.8	15.3	8.7
	5%	WB	2.8	2.9	9.6	13.2	18.6	21.5	23.0	22.2	20.7	16.6	10.6	5.2
		MCDB	4.1	4.6	12.5	17.8	22.6	26.3	27.6	26.3	23.9	19.3	13.3	6.6
	10%	WB	1.3	1.1	6.2	11.0	17.0	20.3	21.9	21.2	19.5	14.6	8.3	3.0
		MCDB	2.2	2.6	9.3	16.4	21.1	24.8	26.4	24.9	23.0	17.5	10.5	4.1
Mean Daily Temperature Range	5% DB	MDBR	8.0	9.5	11.0	12.7	13.0	12.8	13.0	12.7	13.0	11.1	8.7	6.9
		MCDBR	8.9	11.2	15.7	18.3	15.8	15.4	14.7	14.3	15.6	15.2	13.4	9.6
		MCWBR	7.6	8.7	10.8	10.0	8.0	7.3	6.6	7.1	8.5	9.6	9.4	8.2
	5% WB	MCDBR	8.4	10.3	14.1	14.9	13.2	12.7	12.0	11.7	11.8	12.3	11.9	8.8
		MCWBR	7.4	8.5	10.7	9.9	8.0	7.0	6.4	6.6	7.4	9.2	9.3	8.1
Clear-Sky Solar Irradiance	taub	0.287	0.307	0.329	0.360	0.393	0.411	0.403	0.397	0.374	0.339	0.304	0.281	
	taud	2.336	2.278	2.376	2.327	2.289	2.277	2.296	2.313	2.368	2.469	2.507	2.420	
	Ebn at Noon	839	882	911	904	879	861	863	857	850	841	823	818	
	Edh at Noon	80	101	106	120	129	131	127	121	106	84	69	68	
All-Sky Solar Radiation	RadAvg	1.26	2.14	3.56	4.56	5.42	6.04	6.12	5.19	4.15	2.53	1.49	1.00	
	RadStd	0.11	0.24	0.40	0.43	0.53	0.38	0.32	0.32	0.39	0.31	0.20	0.12	